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Dispenser systems

Abstract

A dispenser system for delivering various combinations of dispersants to a body of water with the various combinations of dispersants supplied to the body of water through the interchanging of dispensing cartridges which are attachable to a main dispensing cartridge where a condition for interchanging of dispensing cartridges may be triggered by an event and the dispersants are isolated from a floatation member in the dispenser system.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a divisional of U.S. application Ser. No. 13/999,907 filed Apr. 2, 2014, now U.S. Pat. No. 10,337,200, which claims priority to U.S. Provisional Application No. 61/853,719, filed Apr. 10, 2013, and is a continuation-in-part of U.S. application Ser. No. 13/694,209 filed Nov. 7, 2012, now U.S. Pat. No. 9,039,903, which is a continuation in part of U.S. application Ser. No. 13/573,246 filed Sep. 4, 2012, now U.S. Pat. No. 9,174,177.

STATEMENT REGARDING FEDERALLY SPONSORED RESEARCH OR DEVELOPMENT

(1) None

REFERENCE TO A MICROFICHE APPENDIX

(2) None

BACKGROUND OF THE INVENTION

(3) The concept of floating dispenser for delivering a dispersant to a body of water are known in the art. In one type of floating dispensers the dispensers contain a halogen and float upright in a

body of water until the halogen is consumed whereupon the dispenser flops on its side. In another type of dispenser the dispenser sinks to the bottom of the pool and remains there until the dispersant has been consumed at which time the dispenser floats to the top of the pool to alert the operator to replace the dispenser. In another type of a floating dispenser a nondissolvable weight, such as a marble, is included in the dispenser with the dispenser floating on its side when the dispersant is consumed. As the dispenser floats on its side the marble rolls along the interior of the dispenser and transfers the weight to the end of the dispenser causing the dispenser to invert after the dispersant is consumed.

(4) A typical floating dispenser contains a buoyancy chamber and a dispersant chamber that may contain a dispersant for killing harmful organisms. Once the dispersant is spent the dispenser may be refilled or the dispenser may be replaced with a fresh dispenser. Although the use of dispersants, such as a halogen may be necessary to maintain a body of water such as a pool or spa in a condition where the water can safely be used for recreational purposes there are events or times when additional conditioning of the water may be needed. For example, spring start up of the pool or spa, after heavy use of the pool or spa, after severe weather if the pool or spa is located outdoors, when contaminants build up in the water, when chloramines build up in the water and when a pool is closed for the season are typical events that may require other types of dispersants or water treatment materials to maintain the water in proper condition for recreational use. Oftentimes such water treatment materials are purchased from different sources, which may lead to problems in water maintenance since each of the products may have different directions or instructions on use as well as being incompatible with each other.

SUMMARY OF THE INVENTION

(5) A dispenser system, which preferably is a floating dispenser system, provides a pool or spa owner with multiple water conditioning modes, which may include cartridges that provide a full season conditioning of the body of water to make it suitable for human recreational use such as swimming or soaking. The dispenser system includes a main cartridge containing a dispersant and a first cartridge securable to the main cartridge for floatingly delivering dispersants from both the main cartridge and the first cartridge into a body of water. The dispenser system further includes a set of cartridges containing different dispersants with each of the set of further cartridges interchangeable with the first cartridge to enable the floating dispenser system to provide release of different dispersants or water affecting materials when cartridges are interchanged in response to an event. In one example the floating dispenser floats in an upright condition when a dispersant is present in the main cartridge and the first cartridge and the main cartridge self inverts 180 degrees in response to consumption of the dispersant in the first cartridge to provide a visual alert to replace the spent first cartridge with a fresh cartridge by removing the spent first cartridge from the dispenser and attaching a fresh cartridge to the main cartridge in the floating dispenser.

(6) In another example the floating or immersible dispenser includes an isolator located between the flotation member and the halogen carried in the dispenser with the isolator preventing the dispersible halogen from interfering with the operation of the immersible dispenser.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a front elevation view of a floating dispenser with an upper and lower dispensing cartridge;

(2) FIG. 1B shows the floating dispenser of FIG. 1 with the lower dispensing cartridge replaced by another dispensing cartridge;

(3) FIG. 2 is a side view of the floating dispenser of FIG. 1;

(4) FIG. 3 is a back view of the floating dispenser of FIG. 1;

- (5) FIG. 4 is a bottom view of the floating dispenser of FIG. 1;
- (6) FIG. 5 is a top view of the floating dispenser of FIG. 1;
- (7) FIG. 6 is a side view of the floating dispenser in a body of water;
- (8) FIG. 6A is a side view of the floating dispenser of FIG. 6 as the dispersant in the floating dispenser is partly consumed;
- (9) FIG. 6B is a side view of the floating dispenser of FIG. 6 as more of the dispersant in the floating dispenser has been consumed;
- (10) FIG. 6C is a side view of the floating dispenser of FIG. 6 as the dispersant in the floating dispenser is consumed or substantially consumed;
- (11) FIG. 7 is a top perspective view of the base of a first dispensing cartridge;
- (12) FIG. 7A is a perspective of a support grid of a second dispensing cartridge, which is rotationally secured to the base of the dispenser cartridge of FIG. 7;
- (13) FIG. 7B is a top view of the base of the second dispensing cartridge;
- (14) FIG. 7C is a top view of the base of a lower dispensing cartridge;
- (15) FIG. 7D is a top view of a dispensing cartridge taken along lines 7D-7D of FIG. 1B;
- (16) FIG. 7E is a top view of a dispensing cartridge taken along lines 7E-7E of FIG. 7G;
- (17) FIG. 7F is a side view of a dispensing cartridge;
- (18) FIG. 7G is an exploded view of a bottom dispensing cartridge and a top or main dispensing cartridge in a non engaging condition;
- (19) FIG. 7H shows a front view of the floating dispensing system;
- (20) FIG. 8 is perspective sectional view of the halogen support grid of FIG. 7A showing the location of dispersant pucks and a float with respect to the support grid;
- (21) FIG. 9 is a partial sectional elevation view of the second dispensing cartridge showing the position of the solid dispersant and the float therein;
- (22) FIG. 10 shows a user rotating the two dispenser cartridges with respect to each other in order to select a halogen dispersant rate;
- (23) FIG. 11 shows each of the dispensing cartridges having a different color;
- (24) FIG. 12 is an elevation view of the second dispensing cartridges with a portion of hemispherical cap cut away;
- (25) FIG. 12A is a partial sectional view of a shell with a concealed latch attached thereto; and
- (26) FIG. 13 is a top view of the first dispensing cartridge without the outer hemispherical shell;
- (27) FIG. 14 is a partial sectional view of a floating dispenser revealing a compartment containing a water dispersible material with an isolator; and
- (28) FIG. 15 is a perspective view of the isolator.

DESCRIPTION OF THE PREFERRED EMBODIMENT

- (29) FIG. 1 is a front view of a floating dispenser **10** for use in pools, spas or other bodies of water that require a continuous delivery of a dispersant in order to control a feature or characteristic of the water. A typical example of pools suitable for use of the invention are backyard pop up pools, which hold less than 10,000 gallons of water, although the invention is equally well suited for use in other types of pools or spas.
- (30) Dispenser **10**, as shown in FIG. 1-5, comprises two separable dispensing cartridges, a first upper dispensing cartridge **11** having a general hemispherical shape with an outer cap or shell **11s** (FIG. 12) with a set of protrusions **12** and **13** creating the appearance a frog head and a second lower dispensing cartridge **16** also having a general hemispherical shape with an outer cap or shell **16s** (FIG. 9). The shells of dispenser **11** and **16** are typically made from lightweight polymer plastics such as polypropylene or ABS although other materials may be used without departing from the spirit and scope of the invention. The two dispensing cartridges are assembled into a single dispenser **10** having a general spherical shape.
- (31) The dispensing cartridge **11** and dispensing cartridge are **16** rotationally secured to each other to permit rotation of dispensing cartridge **16** with respect to dispensing cartridge **11** to permit one

to select an appropriate dispensing rate. In the example shown the lower dispensing cartridge **16** comprises a generally hemispherical shaped cap or shell having a stand **18** comprising a flat pole region, which can be used to support the dispenser **10** in a stable upright condition when the dispenser **10** is not in use. Features comprising elongated finger dimples **17** extend midway around the circumference of dispenser cartridge **16** to permit a person to facilitate grasping and holding the dispenser cartridge **16** with one hand while the other hand grasps and rotates dispensing cartridge **11** with respect to cartridge **16** by engaging features such as protrusions **12** and **13**. A reference to FIG. **10** illustrates a user generating a positive rotational grip on the dispenser **10** by placing the fingers of one hand L in the elongated finger dimples **17** of lower dispensing cartridge **16** and the fingers of the other hand R on the features forming the protrusions **12** and **13** to rotate the upper dispenser cartridge **11** with respect to lower dispenser cartridge **16**. The rotation of dispensing cartridge **11** with respect to dispensing cartridge **16** controls the dispersant rate from the dispenser **10** through alignment of ports between a grid **15** (FIG. **7**) of the upper dispensing cartridge **11** and a grid **30** (FIG. **7B**) of the lower dispensing cartridge **16**. Thus, the base **15** and the base **16e** include fluid alignable ports **29** and **34-40** to permit ingress and egress of water between the first cap **11s** and the cap **16s**.

(32) As shown in FIG. **1** the first dispensing cartridge **11** has a general hemispherical shape with an upper protrusion **12** containing an impression **12c** to provide the appearance of a frog eye. Similarly, a second similar protrusion **13** contains a similar identical impression to provide the appearance of a second frog eye. A wavy relief **14** extends across the front of the dispensing cartridge **10** providing a smile to the frog face. In the example shown the dispenser **10** has the appearance of a frog head, however, no limitation thereto is intended as other shapes may be used without departing from the spirit and scope of the invention described herein.

(33) FIG. **2** is a side view of the dispenser **10** and FIG. **3** is a rear view of the dispenser revealing a set of elongated ports **21** for flow of water into and out of dispersant compartments, which are located within dispensing cartridge **11** and dispensing cartridge **16**. In the example shown the elongated ports **21** extend upward from flange **15g** and terminate at a level h, which is normally below the water line when dispenser **10** floats in a body of water, so that a dispersant from ports **21** goes directly into the water. In this example the elongated ports **21** are fixed openings however, if desired the elongated ports may also be provided with variable area openings without departing from the spirit and scope of the invention. A below the water line port is beneficial to water treatment since the dispersant released from the dispenser cartridge **16** goes directly into the body of water.

(34) FIG. **4** shows a bottom view of dispenser **10** with dispensing cartridge **16** having a set of elongated finger dimples **17** extending circumferentially around the dispensing cartridge **16** to permit a user to grasp and support the dispensing cartridge **16** in a users hand. Flange **15g** includes a rectangular shaped opening **15s**, which if desired, can be used to secure a lanyard thereto to enable a user to tether the floating dispenser **10** to a structure and thus maintain the floating dispenser in a fixed location within a body of water.

(35) FIG. **4** reveals the presence of a set of dispensing numbers or dispenser settings **0-8** spaced around the peripheral flange **15g** of dispenser cartridge **11**. Dispenser cartridge **16**, which is rotationally secured to dispenser cartridge **11** includes a U shaped relief **16b** that highlights an arrow **16a** that is shown in alignment with an arrow **15a**, which is located on flange **15g** which is part of dispenser cartridge **11**. The alignment of the two arrows indicates the position where the upper dispensing cartridge **11** may be separated from the lower dispensing cartridge **16** by pulling the dispensing cartridge **11** and dispensing cartridge in opposite directions along a central axis of the dispenser **10**. The separation of the lower dispensing cartridge **16** from the upper dispensing cartridge **11** permits one to replace a spent dispensing cartridges with a fresh cartridge. For example, if the top dispensing cartridge **11** contains minerals, which have a pool life of 6 months, and the lower dispensing cartridge **16** contains a halogen, which is consumed in 3 weeks, after 3

weeks one can separate the spent lower dispensing cartridge **16** from the upper dispensing cartridge **11** and replace the lower spent dispensing cartridge **16** with a fresh dispensing cartridge.

(36) As dispenser **10** floats in a body of water lower dispensing cartridge **16** delivers a dispersant into a body of water through the set of elongated ports **21** in dispensing cartridge **11**. Control of the alignment of a set of internal ports in a grid of each of the dispensing cartridges **11** and **16** allows one to control the rate of dispersant from dispensing cartridge **16**. For example, by rotating dispensing cartridge **11** with respect to dispensing cartridge **16** in a first direction increases the alignment of internal grid ports and consequently the flow area therebetween which increase the dispersant rate from dispensing cartridge **16**. Similarly, rotating dispensing cartridge with respect to dispensing cartridge **16** in an opposite direction decreases the dispersant rate from cartridge **16**. Dispenser cartridge **11** and dispenser cartridge **16**, while separable from each other, are also rotationally secured to each other to permit one to rotate dispenser cartridge **16** with respect to dispenser cartridge **11** thereby increasing or decreasing the flow area through alignment of internal ports in a grid in the dispenser cartridge **11** and a grid in the dispenser cartridge **16** without separating the two dispensing cartridges since there is only one position where the dispensing cartridges can be separated from each other. For example, if a halogen such as chlorine is located in dispensing cartridge **16** the rotation of cartridge **16** with respect to cartridge **11** in one direction increases the alignment of grid ports **35-40** (FIG. 7B) with port **29** (FIG. 7) while an opposite rotation of cartridge **16** with respect to cartridge **11** decreases the alignment of the grid ports **35-40** with port **29**. In either case one alters the volume of water that comes into contact with the chlorine located in dispersing cartridge **16**. In this example, the area of the alignment of flow openings between dispersant compartment in cartridge **16** and cartridge **11** increase as the arrow **16a** is rotated into a position proximate a higher number on the flange **15g** and decreases when rotated in the opposite direction. The increase in the alignment of grid ports allows for a greater dispersant rate of chlorine into the body of water since there are more paths and a greater area for the chlorine to flow from the dispensing cartridge **16** to the body of water.

(37) FIG. 4 shows that dispensing cartridge **16** includes a vent or hole **19**, which is located on the bottom or pole region of dispensing cartridge **16**, and FIG. 5 similarly shows that dispensing cartridge **11** also includes a set of vents or holes **12a**, and **13a**, which are located on the top or pole region of dispensing cartridge **16** and vent holes **12b** and **13b**, which are located on the top protrusions **12** and **13** of dispensing cartridge **16**. The vents permit ingress and egress of air or water to each of the dispenser cartridges depending on the orientation of the dispenser **10** in a body of water. In each case the vents are located at or near the pole regions of dispensing cartridge **11** and dispenser cartridge **16**. During a portion of an operational cycle of the dispenser **10** the vents enable air to escape the dispensing chambers within the dispensing cartridge **11** and **16** and during another portion the vents enable water to enter the dispensing chamber in the dispenser cartridges **11** and **16**. For example, when the floating dispenser **10** is initially placed in a body of water air, which is present in the interior of the floating dispenser, is forced out the top of dispensing cartridge **11** through vents **12a**, **12b**, **13a** and **13b** as water enters the lower vent **19** in dispensing cartridge **16**. This process, which may take a few minutes, allows air to escape from the interior of the dispensing cartridges **11** and **16** as water enters the bottom of dispensing cartridge through vent **19**. Once the dispenser **10** is vented the buoyancy forces of dispenser **10** bring the dispenser to a floating condition as shown in FIG. 6.

(38) FIG. 6 to FIG. 6C illustrate a feature of the dispenser that alerts a person to a condition that the dispersant located in the lower dispensing cartridge **16** has been consumed through an 180 degree inversion of the floating dispenser **10**. The first spherical cap **11** contains a flotation chamber and the second spherical cap **16** may also contain a flotation chamber with a center of buoyancy of the first flotation chamber and the second flotation chamber located along a geometric axis **9** of the dispenser wherein the geometric axis coincides with a vertical flotation axis when the dispersant therein are in an unconsumed state as illustrated by FIG. 6. In the example illustrated in FIG. 6-

FIG. 6C the floating dispenser **10** contains a fixed volume float **29f** (FIG. 12) with the floating dispenser **10** having a bottom heavy state (FIG. 6) when a consumable dispersant **21** is located therein and a top heavy state (FIG. 6C) when the consumable dispersant has been consumed to thereby invert the floating dispenser in an end-to-end condition. A feature of the invention is that the end-to-end inversion may be obtained solely through a weight shift within the dispenser which is due to consumption of the consumable dispersant therein. In this example the fixed volume float **29** comprises a solid float having a specific gravity less than 1 although other types of floats may be used without departing from the spirit and scope of the invention.

(39) In a typical example a first dispersant comprising a batch of minerals, which are located in a dispersant chamber in the top dispenser cartridge **11**, release metal ions into the body of water. The second dispersant, which is located in the lower dispensing cartridge **16**, releases a halogen such as chlorine into the body of water. The dispenser **10** includes sufficient flotation to maintain the dispenser **10** in a floating condition even though the dispersants located in dispensing cartridge **11** and dispensing cartridge **16** have a specific gravity greater than 1. The dispenser **10** is initially configured so that the weight of the dispersant in the top dispensing cartridge **11** and the top dispensing cartridge is less than the weight of the dispersant material in the bottom dispensing cartridge **16** and the bottom dispensing cartridge which caused the center of gravity of dispenser **10** to be located below a geometric center of dispenser **10**. In this condition the dispenser **10** floats in an upright condition as illustrated in FIG. 6 with a substantial portion of the dispenser **10** located below the water line **50**. FIG. 6 also shows the vertical flotation axis **9** in alignment with a geometric axis of the dispenser **10** with respect to a water line **50**.

(40) As the dispersant in the bottom dispenser cartridge **16** is gradually consumed, i.e. dissipated into the body of water, the weight of the dispersant in a dispensing chamber in dispenser cartridge **16** decreases which causes the geometric axis **19** of dispenser **10** to form a slight angle to the vertical as shown in FIG. 6A. In this condition the water line **50** is still above the elongated ports **11** so that any dispersant released therefrom goes directly into the water. On the other hand the release of metal ions from the top cartridge **11** has relatively little effect on the weight of material in the top dispensing cartridge and hence the floating orientation of the dispenser. Consequently, the shifting of weight balance between the top dispensing cartridge **11** and the lower dispensing cartridge **16** generates a torque causing the lower dispensing cartridge **16** to move to the condition shown in FIG. 6B where the weight of the top dispensing cartridge **11** and the lower dispensing cartridge **16** are in substantial equilibrium with the geometric axis **9** located substantially parallel to the water line **50**.

(41) As the dispensing material in dispensing cartridge becomes further consumed the weight of the lower dispensing cartridge **16** becomes less than the weight of the upper dispensing cartridge **11** causing the dispenser **16** to invert to the condition shown in FIG. 6C. In the inverted condition the protrusions **12** and **13** in the dispenser **10** are located below the water line **50** as the upper dispensing cartridge **11** is located below the water line **50**. Since a portion of the weight of the dispenser **10** has been consumed the dispenser **10** also floats higher in the water as indicated by the waterline **50**. In the inverted condition the bottom of dispensing cartridge **16** is visible above the water line **50** to form a visual indication that the dispensing cartridge **16** needs to be replaced.

(42) In the example shown the use of vent hole **19** and vent holes **12a**, **12b**, **13a** and **13b** (FIG. 4 and FIG. 5) allow air or water trapped inside the dispenser **10** to come into equilibrium state as the dispenser **10** rotates from a full condition (FIG. 6) to the empty condition or the inverted condition (FIG. 6C).

(43) While the reversal of the orientation of the dispensing cartridges **11** and **16** in floating dispenser **10** may be used to indicate the dispersant in the lower dispensing cartridge is spent one may also enhance the recognition of the lack of dispersant through use of a different color for each of the dispenser cartridges. FIG. 11 illustrates this feature with the lower dispensing cartridge **16** having a blue color and the top dispensing cartridge **11** having a green color. Thus, the portion of

the dispenser above the water line changes from green to blue providing a visual alert to the operator that the dispensing cartridge has been consumed and needs to be replaced. In addition, the position of a hemispherical shell without protrusions located above the water line forms another visual indicator that the dispersant in dispensing cartridge **16** has been spent. While blue and green have been selected other contrasting colors may be used without departing from the spirit and scope of the invention.

(44) FIG. 7 shows a perspective view of dispensing cartridge **11** revealing a flat grid or base **15** having a flange **15g**, which extends from side to side of the dispensing cartridge **11** to form an equatorial closure for the open end of the hemispherical shaped dispenser **11**. Base **15** extends slightly beyond the hemispherical shell **11s** to form a flange **15g** for holding reference information to enable an operator to select the proper dispensing rates.

(45) Grid **15** includes a set of three peg like protrusions **15d** that extend outward from the face of grid **15** to engage edges of circumferential recesses **41a**, **41** (FIG. 7B) located in the face of grid **30** on the dispensing cartridge **11**. When the protrusions **15d** are in engagement with the edge of the recesses **41a**, **41** located in the face of grid on dispensing cartridge **16** it becomes difficult to rotate dispensing cartridge **11** with respect to dispensing cartridge **16**. However, since the lateral extensions **20b** and **20c**, which hold grids on dispensing cartridge **15** and dispensing cartridge **16** in proximity to each other can flex slightly one can overcome the resistance through application of hand torque to dispensing cartridge **11** with respect to dispensing cartridge **16**. The torsional resistance to rotation of dispensing cartridge **16** with respect to dispensing cartridge **11** provided by protrusions **15d** hinders the accidental rotation of the cartridges with respect to one another but ensures that once a dispensing setting has been selected the dispensing setting remains until a person deliberately changes the setting by again rotating the dispensing cartridge **11** with respect to dispensing cartridge **16** through as illustrated in FIG. 10.

(46) FIG. 7 shows that grid **15** includes a triangular shaped port **15a** that can be rotationally aligned with each of the set of ports **35-40** in a base **16e** in dispensing cartridge **16**. In this example, the fluid ports **21** in the top dispensing cartridge **11** are visible as well as a solid float **29** having a specific gravity less than one, which provides buoyancy to dispensing cartridge **11**. Although a solid float is shown other types of floats may be used including closed air compartments.

Extending upward from the center of base **15** is a connector **20** having a fixed central cylindrical post **20a** that permits rotation of dispensing cartridge **11** with respect to dispensing cartridge **16** by serving as a pivot pin. Extending outward in one direction from post **20a** as well as spaced from the surface of grid **15** is a first lateral extension **20b**. Similarly, extending outward in the opposite direction from post **20a** as well as spaced from the surface of grid **15** is a second lateral extension **20c** which has a different shape from the first lateral extension so that there is only one position where the dispensing cartridges **11** and **16** can be joined in face to face contact. The lateral extensions **20c** and **20b** maintain the grid or base **15** of dispensing cartridge **11** and the grid **30** of dispensing cartridge **16** in face to face contact with each other as illustrated in FIG. 7A, which shows an isolated view of the grid **30** of dispensing cartridge **16**. Grid **30** includes a set of radial ribs **30R** and a set of circumferential ribs **30C** that provide rigidity to grid **15** to maintain the grid **30** in face to face contact with grid or base **15** while also forming a portion of a halogen compartment. Visible within the radial and circumferential rings are a set of fluid ports **40**, **39**, **38**, **37**, **36** and **35**. A reference to FIG. 7B shows the opposite face of grid **30** revealing the set of fluid ports **35**, **36**, **37**, **38**, **39** and **40** that are located around the opening **34**. Opening **34** has the shape of post **20** and lateral extensions **20a** and **20c** to permit the placement of grid **30** in face to face contact with grid **15** as the post **20** and lateral extensions **20b** and **20c** are inserted through the opening **34** to engage the opposite face of grid **30**. Once in position a slight rotation of grid **15** with respect to grid **30** causes the lateral extension **20b** and **20c** to hold the grids in face to face contact as illustrated in FIG. 7A. Since the lateral extensions **20b** and **20c** are of different shape there is only one orientation of dispensing cartridge **11** with respect to dispensing cartridge **16** where the

dispensing cartridge **11** and dispensing cartridge **16** can be attached or separated from each other.

(47) Extending upward from grid **30** are a set of four resilient extensions or tabs **30a**, which have an opening therein for engaging a latch. FIG. **12A** shows a concealed latch **30d** having the shape of a hook, which is located on the inside of the hemispherical shell **11s** of dispensing cartridge **11**, that can be brought into engagement with a concealed tab **30a** to hold the shell **11s** and base **30** in an assembled condition. That is, by forcing the dispensing cartridge **11** onto the grid or base **30** the latch **30d** (FIG. **12A**) in the shape of a hook extends into the opening in extension **30a** to hold the shell **11s** and the grid **30** proximate each other. In the example shown a set of four latches is used however, more or less latches could be used. Since the latches and the extensions are located internal to dispensing cartridge **11** and thus hidden from view it inhibits persons such as children from accidentally opening the dispensing cartridge **11** and coming into direct contact with the dispersants therein. A similar or identical set of latches and extension can be found in dispenser shell **16** and base **15** to prevent consumers coming into accidental contact with the contents therein.

(48) FIG. **7B** shows that grid **15** includes a set of arc like openings **41** and **41a** for receiving the extensions **15d** shown in FIG. **7**. Since the grid **30** must be displaced in an axial direction to disengage the extension **15d** with the sides of the openings **41** and **41a** the coaction of the extensions and the openings **41** and **41a** hinder the ability of dispensing cartridge **11** to rotate with respect to dispensing cartridge **16** unless the user can overcome the rotational resistance through flexing of the grids **30** and **15**. As the shells and grids are typically made from a polymer plastic which permits a small amount of flexing one can overpower the force of engagement between face **15** and face **30** through hand torque as illustrated in FIG. **10**.

(49) Grid **15** as shown in FIG. **7B** includes an annular groove **31** with a set of openings **32** that provide fluid communication between openings **15e** (FIG. **7**) in grid **15** to allow air to escape the dispersant chamber in dispensing cartridge **16**. Typically, as the dispersant in the dispensing cartridge **16** dissipates water enters the dispensing cartridge **16**. When the dispensing material reaches a consumed state the lower dispensing cartridge **16** becomes more buoyant than the top dispensing cartridge **11** and the geometric axis **9** begins to wander from vertical. As the process of reversing of dispensing cartridges **11** and **16** begins the openings **15e** which are alignment with the channel **31** and the ports **32** together with the vents **19**, **13a**, **13b**, **12a** and **12b** allow air to escape dispenser **10** as the dispenser **10** goes through the inversion phase illustrated in FIG. **6** to FIG. **6C**.

(50) FIG. **1B** shows the floating dispenser **10** of FIG. **1** with the lower dispensing cartridge **16** replaced by another dispensing cartridge **116**.

(51) FIG. **7C** shows a partial sectional top view of lower dispensing cartridge **116**, which is similar to top dispensing cartridge **11** with identical parts having the same numbers. The cylindrical base **116e** is similar to base **16** with base **116e** containing a set of elongated radial ports **135**, **137**, **138**, **139** and **140** for water to obtain access to the three dispersant compartments **101**, **102** and **107** which are formed by a set of dividers **103**, **105** and **104** that join at the center of the cartridge to form three separate dispensing compartment **101**, **102**, and **107** in cartridge **116e**.

(52) FIG. **7D** shows a view taken along lines **7D-7D** of FIG. **1B** revealing the three dispersant chambers **102**, **107** and **101**. In the example shown the dispersant compartment **102** contains a dispersant **110**, the dispersant compartment **107** contains a dispersant **112** and the dispersant compartment **101** contains a dispersant **111**. For example, one compartment may contain a clarifier, another may contain an algicide and another may contain a phosphate remover. Other dispersants may include stain removers although other dispersants may be used without departing from the spirit and scope of the invention.

(53) FIG. **7G** shows an exploded view of the floating dispenser top or main dispensing cartridge **11** and a first dispensing cartridge **120** for attachment thereto. In the example shown the top or main dispensing cartridge **11** includes a connector or attachment member **20** having wings or lateral extensions **20b** and **20c** for securing the first dispensing cartridge **120** to the main or top dispensing cartridge **11**. The dispensing cartridge **120** contains a base **121** and a set of elongated finger

dimples **122** extending midway around the circumference of dispenser cartridge **120** to permit a person to facilitate grasping and holding the dispenser cartridge **120** with one hand while the other hand grasps and rotates dispensing cartridge **11** with respect to cartridge **120** by engaging features such as protrusions **12** and **13** to thereby secure the lower dispensing cartridge **120** to the upper dispensing cartridges **11** through base to base engagement of the base of the dispensing cartridge **120** to the base of the main dispensing cartridge **11**.

(54) FIG. 7E shows a top view of a cartridge dispenser **120** taken along lines 7E-7E of FIG. 7G. In this example cartridge dispenser **120** contains a single dispersant compartment **125** with a dispersant **126** therein. For example, a shock dispersant such as dichloroisocyanuric, commonly referred to as dichlor, may be used in dispenser cartridge **120**.

(55) FIG. 7F shows a front view partially in section of a further dispenser cartridge **130** having a first compartment **133** and a second compartment **134** for receiving a dispersant. In this example the cartridge dispenser **130** may include a base **132** and a set of elongated finger slots **131** to enable securement of the dispenser cartridge **130** to the first dispensing cartridge **11** through engagement of top base **136** with the attachment members **20c**, **20d** of dispensing cartridge **11**.

(56) FIG. 7H shows a dispenser system **150**, which is adaptable for full season use in an outdoor pool or spa with the system comprising a main dispensing cartridge **11** having a base **15** (FIG. 7), a float **29f** (FIG. 12), and a dispersant chamber **11e** (FIG. 12) containing a first dispersant **42** (FIG. 12) for water release therefrom with base **15** having an attachment member **20** thereon comprising cylindrical post **20** and lateral extensions **20c** and **20b** (FIG. 7). In this example, the dispenser cartridge **11** can be secured to each of the dispensing cartridges **16**, **116**, **120** and **130** with each of the dispensing cartridges **16**, **116**, **120** and **130** interchangeable with each other and maintainable in a dispensing mode when attached to the top or main dispensing cartridge **11**.

(57) The dispenser system **150** may include a set of dispensing cartridges that are prefilled with each of the dispensing cartridges releasably securable and interchangeable with each other for attachment to first dispensing cartridge, where the first dispensing cartridge may contain a dispersant that one wants to maintain in the body of water at all times. For example, a source of metal ions that kills harmful organisms. One such source of metal ions may be found in the mineral pack sold by King Technology of Hopkins Minnesota.

(58) A second dispensing cartridge **116** (also shown in FIG. 7C and FIG. 7D) having a base **116e** releasably securable to the attachment member **20** of the first dispensing cartridge **116** with the second dispensing cartridge **116** containing three separate dispersant **111**, **110**, or **112** (FIG. 7D) for water release therefrom to thereby enable the first dispensing cartridge **11** and the second dispensing cartridge **116** to simultaneously supply dispersants to the outdoor pool or spa.

(59) Dispenser system **150** includes a third dispensing cartridge **120** with the third dispensing cartridge containing a third dispersant **125** for water release with the third dispensing cartridge **120** interchangeable with dispensing cartridge **16** to thereby enable the dispensing cartridge **11** and the third dispensing cartridge **120** to simultaneously supply dispersant to the outdoor pool or spa when the third dispensing cartridge is interchanged with the second dispensing cartridge.

(60) Dispenser systems **150** may include a fourth dispensing cartridge **130** with the fourth dispensing cartridge **130** containing a further dispersant **134a** in compartment **134** and a further dispersant **133a** in dispersant compartment **133** for water release from the fourth dispensing cartridge **130** interchangeable with dispensing cartridge **16** to thereby enable the fourth dispensing cartridge **130** and the first dispensing cartridge **11** to simultaneously supply dispersant to the outdoor pool or spa when the fourth dispensing cartridge is interchanged with the second dispensing cartridge **16**.

(61) In the dispenser system **150** the various dispensing cartridges include chambers for carrying or delivering water dispersant to a body of water. For example, the main dispensing cartridge **11** may contain a source of silver ions to enable silver ions to be deliverable to the outdoor pool or spa while dispersants are being supplied to the outdoor pool or spa from a dispensing cartridge attached

to the main dispensing cartridge **11**. The first dispensing cartridge **16** may contain a halogen, the second dispensing cartridge **116** may contain an algaecide, the third dispensing cartridge **120** may contain a shock dispersant and the fourth dispensing cartridge **130** may contain a different water treatment material. A shock dispersant usable in the system **150** may include, although no limitation thereto is intended, one or more of the following monopersulfate, dichloroisocyanuric, soda ash or bicarbonate with a weak acid, dichloroisocyanuric with or without a clarifier and sodium bromide.

(62) The dispensing system **150** may also contain a dispersant cartridge having a dispersant for affecting a pH of the body of water. For example one dispenser cartridge may have soda ash for increasing pH and another dispensing cartridge may have citric acid or muriatic acid for decreasing pH. In a multiple compartment dispensing cartridge including three compartments one of the compartments may contain a clarifier, a second compartment may contain sodium bicarbonate citric acid and the third compartment may contain an algaecide or a phosphate remover.

(63) In the example shown in FIG. 7H the dispensing cartridges are prefilled to avoid the consumer having to come into contact with the dispersants, however, if desired the one or more of the dispensing cartridges may be consumer refillable.

(64) The invention also includes a method of delivering multiple dispersants from a single floating dispensing platform **11** by supplying a first water dispersant to a top or main dispensing cartridge having a cartridge attachment member, supplying a second water dispersant to a first dispensing cartridge having a cartridge attachment member and supplying a third water dispersant to a second dispensing cartridge having a cartridge attachment member. One can then attach the main dispensing cartridge to the first dispensing cartridge, place the main dispensing cartridge and the second dispensing cartridge in a body of water. To initiate the next step with the water treatment system **150** one removes the main dispensing cartridge and the first dispensing cartridge from the body of water and replaces the first dispensing cartridge with the second dispensing cartridge; and places the main dispensing cartridge and the second dispensing cartridge into the body of water to enable the dispenser system **150** to deliver dispersants from the main dispensing cartridge and the second dispensing cartridge. Typically, the main dispersant comprises a source of silver ions and either the second water dispersant or the third water dispersant may be either a shock dispersant or an algaecide.

(65) The system **150** further provides a method of delivering multiple dispersants during different time periods based on the occurrence of an event. For example, the event may be spring start up of the pool or spa, after heavy use of the pool or spa, after severe weather if the pool or spa is located outdoors, when containments build up in the water, when chloramines build up in the water and when a pool is closed for the season owner are typical conditions that may require other types of dispersants to maintain the water in proper condition for recreational use. By conditioning as used herein it is understood to mean the use of dispersant to bring the pool or spa water into a state where the water is suitable and safe for a person to immerse his or her body therein. Such bodies of water may be referred to as recreational bodies of water. Some of the various dispersants, which may be used have been described herein although other dispersants may be used.

(66) In the method of delivering multiple dispersants to a body of water one attaches a first dispensing cartridge to a main dispensing cartridge where the first dispensing cartridge contains a first dispersant and the main dispensing cartridge containing a second dispersant. One can then place the main dispensing cartridge and the first dispensing cartridge in a body of water to condition the water by delivering dispersants from the main dispensing cartridge and the first dispensing cartridge into the body of water. After the occurrence of an event one can remove the main dispensing cartridge and the first dispensing cartridge from the body of water and replace the first dispensing cartridge with a further dispensing cartridge having a further dispersant. In the next step one places the first dispensing cartridge and the further dispensing cartridge into the body of

water to deliver the first dispersant and the further dispersant into the body of water to condition the water with different dispersants. Similarly, the second dispensing cartridge may be replaced by a third dispensing cartridge to provide further delivery of dispersants to a body of water.

(67) While various dispersants that can affect the water characteristics have been described and outdoor pools and spas have been described the invention is not limited to use therein and may be used in other bodies of water including those bodies of water held in backyard inflatable pools or pop up pools. Likewise other dispersants than those described herein may be used in the dispenser system described herein without departing from the spirit and scope of the invention.

(68) While the invention has been described in regard to outdoor pools and spas the invention may be used in other water applications, for example, indoor pools and spas, ponds, fountains, water holding tanks that may or may not be recirculating.

(69) FIG. 4 show dispensing cartridge **16** having a vent **19** and FIG. 7B shows a connector **34** for attachment to a flotation member **11** for supporting the dispensing cartridge **16** and a consumable dispersant **21** (FIG. 9) in a body of water when the vent **12a**, **12b**, **13a** and **13b** in the flotation member **11** is located above a water line or the vent **19** in the dispensing cartridge is located below the water line. The dispensing cartridge **16** and the consumable dispersant **21** having a specific gravity greater than one when the consumable dispersant is in an unconsumed state and a specific gravity of less than one when the consumable dispersant has been consumed therein thereby causing the dispensing cartridge **16** to become more buoyant than the floatation member **11** as the vent **12a**, **12b**, **13a** and **13b** in the flotation member is brought below the water line as illustrated in FIG. 6C and the vent **19** in the dispensing cartridge is brought above the water line.

(70) Thus the examples show the dispensing cartridge **16** including a flotation member **11** (FIG. 12) having a fixed volume float **29f** and a connector **20** (FIG. 7) securing the flotation member **11** with fixed volume float **29f** to the dispensing cartridge **16** with the fixed volume float and the dispensing cartridge having a center of mass such that the dispensing cartridge normally floats below the flotation member (FIG. 6) when the consumable dispersant is an unconsumed state and the dispensing cartridge floats above the floatation member (FIG. 6C) when the consumable dispersant has been consumed to thereby provide an indication that the consumable dispersant should be replaced.

(71) FIG. 8 shows a perspective view of dispenser **10** with the hemispherical shell **16s** of dispensing cartridge **16** removed to reveal a solid dispersant **21** in the form of two pucks of chlorine and a float **22** having a center of flotation located on a geometrical axis **9** of the dispensing cartridge. Similarly, the dispersant, which may be a single halogen puck or multiple halogen pucks such as chlorine or bromine is centrally located along axis **9**. As can be seen from FIG. 8 one side of puck **21** is supported by the grid **30**. In the example shown two cylindrical halogen pucks **21** are shown, however, one as well as multiple pucks may be used. Other forms of halogen may be used, for example, tablets or granules as the dispersant in dispensing cartridge **16**. Similarly, other dispersants besides halogens may be used in dispensing cartridge **16** without departing from the spirit and scope of the invention.

(72) The floating dispenser may be sold as an assembled unit with an upper cartridge **11** and a lower cartridge **16** in an assembled condition with the amount of dispersant in the upper cartridge **11** having a pool life equal to the pool life of the dispersant in the lower dispensing cartridge **16**. In still other cases one may use dispersants in the upper cartridge **11** that have a longer pool life than the dispersants in the lower pool cartridge. For example, the dispersant in the upper pool cartridge may have a pool life of three times the pool life in the lower dispensing cartridge. In such cases the floating dispenser may be sold with two spare lower dispersant cartridges that can be replaced. In use in outdoor pools, which have seasonal use one may have the floating dispenser with the dispersant in the upper chamber sufficient to last a season while the consumable dispersant in the lower dispersant cartridge is replaced with a fresh cartridge based on the consumption of the dispersant in the lower dispensing cartridge.

(73) FIG. 9 shows a cutaway view showing the pucks **21** confined lateral by a circumferential rib **16c** and axially by a axial ribs **16c** and **16d**, which together with the grid **30** form a dispersant compartment for the halogen pucks **21**. Located below the halogen pucks **21** is a float **22** to provide additional buoyancy to the bottom member. Float **22** may or may not be used depending on the buoyancy relation between the materials in the top dispensing cartridge **11** and the bottom dispensing cartridge. In the example shown the bottom float **22** has the center of buoyancy CB located on the geometrical axis **9**.

(74) FIG. 12 shows a side view of dispenser **10** with the hemispherical shell **11s** of dispenser **11** cutaway to reveal float **29f** and the dispersant **42** located therein. FIG. 13 shows a top view of the dispenser **10** without the hemispherical shell of dispensing cartridge **11** revealing a bow tie shaped container with porous walls with water treatment materials contained therein. In this example the dispersant **42** contains a source of metal ions that releases silver ions into the water to assist the halogen in ridding the water of harmful organisms. An example of a source of metal ions is shown and described in King U.S. Pat. Nos. 6,527,952; 6,652,811 and 7,014,780, which are hereby incorporated by reference. A feature of the source of metal ions is that while the metal ions are released the weight of the source of metal ions remains relatively constant. However, the halogen dispersant, which has a specific gravity in excess of 1, is gradually consumed thus causing the center of gravity of the dispenser **20** to change between the lower dispensing cartridge **16** which weighs less as a result of the dispersant consumption while the upper dispensing cartridge **11** remains at substantial the same weight. The shift in the center of gravity generates a torque that causes the dispenser **10** to rotate from the upright condition shown in FIG. 6 to the inverted position shown in FIG. 6C.

(75) FIG. 13 shows a cutaway top view of dispenser **11** with shell **11s** removed to reveal a porous container **40d** having a bow tie shape holding a source of metal ions **9**. Located below container **40d** is a float **29f** that provides flotation to dispensing cartridge **11**. The center of gravity of porous container **40d** and the contents **9** may be located on the vertical geometric axis **9** or if desired the center of gravity may be located radially to one side of the geometric axis in order to provide a flotation bias during the rotation of the dispenser **10** from the full to the empty condition as illustrated in FIGS. 6 to 6C. For example, offsetting the center of gravity of porous container and the contents toward the portion of the dispensing cartridge containing the elongated ports **20** causes the elongated ports to remain under water as the dispenser **10** rotates from the full to the empty condition. In such a case mineral chamber **40d** is asymmetric with respect to a central axis **9** of the dispenser **10**.

(76) FIG. 14 shows a cutaway view of a two part floating dispenser or immersion dispenser with the bottom dispenser **16** including a flotation member **22** and a set of halogen pucks **21**, which are laterally confined by circumferential ribs **16c**. The ribs form a dispersant compartment for laterally restraining halogen pucks **21**. Located below the halogen pucks **21** and at the bottom of dispenser **16** is the flotation member **22**, which in this example is made of lightweight foam to provide additional buoyancy to the bottom dispenser **15** of the two floating dispensers **15** and **16**, which are secured to each other. In this example a first float **29f** (FIG. 12) is located in top dispenser **11** and a second float **22** is located at the bottom of the floating dispenser **16** with the float **22** separated from halogen pucks **21** by an isolator **200**. While flotation member **22** is insufficient to support both dispensers **15** and **16** the flotation member **22** aids in controlling the flotation orientation of the floating dispensers **15** and **16** thus enabling the dispensers **15** and **16** to change the flotation orientation as the dispersant in dispenser **16** is dispersed into the body of water. Isolator **200** comprises a cylindrical member that separates the stack of halogen pucks **21** from the flotation member **22**. In this example flotation member **22** comprises a closed cell plastic foam such as polyethylene although other foam flotation materials may be used without departing from the spirit and scope of the invention. As the halogen puck **21** dispenses or dissolves in the presence of water in the dispenser **16** the halogen puck changes in shape and size. Since a halogen puck **21** in reduced

size may accidentally come into contact with the flotation member **22** and adhere thereto during a halogen-dispensing phase the isolator **200** is placed between the halogen **21** and the float **22**. The isolator **200** prevents accidental contact or adherence of the puck **21** to the float **22** during the dissolution of the puck **21** since the adherence of a partially consumed halogen puck to the float may disrupt the flotation balance or flotation orientation of the floating dispenser **15** and floating dispenser **16** thus providing an erroneous signal to an observer.

(77) An example of a suitable material for an isolator is a polyester, preferably in the form of a rigid disk that extends substantially coextensive with the halogen puck **21** to prevent the halogen **21** from accidentally adhering to the foam of float **22** as the halogen is consumed within the dispenser. Preferably member **200** may be a rigid disk of polyester i.e. for example Mylar (trade name) as shown in FIG. **9A** with the disk having a diameter **D** which is sufficiently large so that it prevents the halogen puck **21** in a partially dissolved state from accidentally contacting and adhering to the foam float **22** since the adherence to the float may affect the flotation balance or orientation of the dispenser. Although polyester has been described as a suitable material for an isolator other materials may be used without departing from the spirit and scope of the invention. While the isolator disk is shown in use with dispensers **15** and **16** that float as a unit the isolator disk may also be used with other immersible dispensers where the halogen is located below the water line and the immersible dispenser may respond by either floating or sinking in a body of water in response to a change in the dispersant in one or both of the dispensers. In the example shown the foam float **22** is spaced from the isolator **200** however, the isolator may have one side in contact with the float and the other side in contact with the halogen puck.

(78) While the floating dispenser has been described with respect to two dispersants the floating dispenser may be used with only a single dispersant. For example, floating dispenser **10** contains a fixed volume float **29** and a bottom heavy state when a consumable dispersant is located in the floating dispenser **10**, which causes the floating dispenser to float in a first orientation as illustrated in FIG. **6**. When the consumable dispersant has been consumed, the floating dispenser **10** inverts end-for-end from the first orientation to a second orientation, as shown in FIG. **6C**, due to the floating dispenser changing from a bottom heavy state to a top heavy state caused solely by the consumption of the consumable dispersant in the floating dispenser.

(79) Thus the invention also includes a method of treating a body of water, for example sanitizing the water, comprising the steps of attaching a first hemispherical shaped dispenser **11** having a first dispersant therein to a second hemispherical shaped dispenser **16** having a second dispersant therein where the second dispenser **16** with the second dispersant is less buoyant than the first dispenser. One positions the first dispenser **11** with respect to the second dispenser **16** to select a dispersant rate of the second dispersant (FIG. **10**). Next, one places the first dispenser **11** with the second dispenser **16** attached thereto in the body of water to begin the process of water treatment as illustrated in FIG. **6**-FIG. **6C**.

(80) One can remove the second dispenser **16** from the first dispenser **11** from the body of water when the second dispenser **16** becomes more buoyant than the first dispenser. One can then replace the second dispenser **11** with a further dispenser and place the first dispenser **16** and the further dispenser into the body of water to continue the sanitizing of the body of water. This works well when the first dispersant is a long term pool treatment dispersant, such as a source of ions, and the second dispersant is a consumable dispersant i.e. a halogen having a shorter pool life than the dispersant in the first dispenser; however, other dispersants may also be used in delivering materials to the body of water.

(81) The method may include the step of securing either the first dispenser or the second dispenser to a pool structure with a lanyard to restrain the first dispenser and the second dispenser from movement about the body of water.

(82) The method further includes the step of grasping the first dispenser in one hand and the second dispenser in the other hand and rotating the first dispenser with respect to the second dispenser to

select a dispersant rate by increasing or decreasing the fluid ports between the first dispenser and the second dispenser as illustrated in FIG. 10, FIG. 7B and FIG. 7C.

Claims

1. An immersible dispenser comprising: a dispensing cartridge attachable to a further dispensing cartridge with the dispensing cartridge having a hemispherical shape with a vent therein and a top base with a grid having an opening for attachment to the further dispensing cartridge with a set of fluid openings therein for ingress and egress of water therethrough while the vent allows air to escape from a dispensing chamber in the further dispensing cartridge when the dispensing cartridge is attached to the further dispensing cartridge; a water dispersible material comprising a halogen puck located in the further dispensing cartridge; a fixed volume float located within the dispensing cartridge to provide flotation to the dispensing cartridge; a flotation member comprising a closed cell polyethylene foam located below the water dispersible material when the dispensing cartridge and the further dispensing cartridge are secured to each other, said flotation member insufficient to support the dispensing cartridge and the further dispensing cartridge in a body of water but said flotation member aids in controlling a flotation orientation of the dispensing cartridge and the further dispensing cartridge to enable the dispensing cartridge and the further dispensing cartridge to change the flotation orientation as the dispersant in the further dispensing cartridge is dispersed into a body of water; and an isolator having a top face in contact with the water dispersible material with the isolator located within the further dispensing cartridge and between the flotation member and the water dispersible material, said isolator laterally confined within the further dispensing cartridge with the isolator preventing the flotation member and the halogen puck from coming into contact with each other during a dissipation of the halogen puck, thereby preventing adherence of the water dispersible material to the flotation member during a water dissolution of the water dispersible material and a disruption of a flotation balance of the immersible dispenser.
2. The immersible dispenser of claim 1 wherein the flotation member and the water dispersible material are located below a water line as the water dispersible material dissipates into the body of water and the flotation member comprises a closed cell foam and the isolator comprises a rigid disk preventing the flotation member and the halogen puck from coming into contact with each other.
3. The immersible dispenser of claim 1 wherein the isolator comprises a layer of polyester film having a thickness of about 0.005 inches.
4. The immersible dispenser of claim 1 with the water dispersible material located in a bottom of the further dispensing cartridge causing the immersible dispenser to float in a first orientation when the water dispersible material, the isolator, and the flotation member are located below the water line and in a second orientation after the water dispersible material has dissipated into the body of water.
5. The immersible dispenser of claim 1 wherein the isolator comprises a circular disk having a diameter substantially equal to a diameter of the halogen puck, and the further dispensing cartridge includes a stack of halogen pucks with the isolator face circular disk having a face in contact with a face of only one of the stack of halogen pucks.
6. An immersible dispenser comprising: a first dispensing cartridge having a hemispherical shape, the first dispensing cartridge including a vent therein and a first base, wherein the first base includes a first set of alignable fluid access ports; a fixed volume float located within the first dispensing cartridge to provide flotation to the first dispensing cartridge; a second dispensing cartridge configured to be rotatably secured to the first dispensing cartridge, the second dispensing cartridge including a second set of alignable fluid access ports; a water dispersible material positioned within the second dispensing cartridge; a flotation member positioned within the second dispensing cartridge and below the water dispersible material when the first dispensing cartridge and the second dispensing cartridge are secured to each other and within a first flotation

orientation; and wherein rotation of the first dispensing cartridge relative to the second dispensing cartridge controls a dispersant rate of the water dispersible material through a change in alignment of the first set of alignable fluid access ports relative to the second set of alignable fluid access ports, whereby rotation from an unaligned position towards an aligned position increases the dispersant rate permitting an ingress and an egress of water therebetween.

7. The immersible dispenser of claim 6, wherein the water dispersible material includes a halogen puck.

8. The immersible dispenser of claim 6, wherein the flotation member is insufficient to support the immersible dispenser in a body of water, the flotation member configured to aid in controlling a flotation orientation of the immersible dispenser to change the flotation orientation as the dispersant in the second dispensing cartridge is dispersed into the body of water.

9. The immersible dispenser of claim 6, wherein the flotation member comprises a closed cell polyethylene foam.

10. The immersible dispenser of claim 6, wherein when the water dispersible material within the second dispensing cartridge dissipates into a body of water, the first dispensing cartridge and the second dispensing cartridge move from the first flotation orientation to a second flotation orientation.
